Smart Pantry Manager

A Java Android app that helps reduce food waste. The user records the ingredients they already have at home (their pantry), and the app suggests only the recipes they can cook **right now**, with no shopping trip needed.

Built for the Mobile App Development 700 practical assignment.

The strict-matching rule

A recipe is shown as "suggested" only if **every** ingredient it needs is in the pantry, in at least the required quantity.

- A recipe that needs 5 ingredients when the pantry has 4 of them never appears in the main list.
- Names are matched in a forgiving way, so "Tomatoes", " tomato " and "tomato" all count as the same ingredient.
- Units are converted, so 0.06 kg of cheese satisfies a recipe that needs 60 g.
- Optional extra: a separate **Almost Their** list shows recipes missing exactly one ingredient. It is kept apart from the strict list.

Features

- Add, edit and delete pantry items (name, quantity, unit, optional expiry date) with input validation
- Pantry list with search and sort (A-Z or expiring first)
- 18 pre-loaded recipes, each with ingredients and simple steps
- Suggested Recipes screen using the strict-matching rule, with a clear message when nothing matches
- Recipe detail screen, plus an "I cooked this" button that removes the used ingredients from the pantry
- Settings screen with a toggle that highlights items expiring within 3 days
- Bottom navigation bar to move between the main screens

Screens

Screen	Class
Pantry List	MainActivity
Add / Edit Ingredient	AddEditIngredientActivity
Suggested Recipes	SuggestedRecipesActivity
Recipe Detail	RecipeDetailActivity
Settings	SettingsActivity

Database choice: SQLite

I chose **SQLite** (using SQLiteOpenHelper) because:

- It is built into Android and stores data locally on the device, so the app needs no internet connection and no account.
- It is the approach covered in the module's persistent data chapter.
- It keeps the design easy to explain from end to end, and the data is still there after the app is closed and reopened.

The database has three tables:

Table	Columns
pantry	id, name, quantity, unit, expiry
recipes	id, name, steps
recipe_ingredients	id, recipe_id, name, quantity, unit

The recipes are inserted automatically the first time the app runs.

Tech used

- Java (not Kotlin)
- Android Studio
- SQLite via SQLiteOpenHelper
- ListView with a custom BaseAdapter
- Intents for moving between screens and passing data
- Material bottom navigation

How to set up and run

1. Install Android Studio.
2. Clone this repository:

```
git clone https://github.com/tanakachimunhu56-bit/SmartPantryManager
```

3. In Android Studio, choose **File > Open** and select the cloned project folder.
4. Wait for **Gradle sync** to finish (click **Sync Now** if a banner appears).
5. Create or start an emulator in **Tools > Device Manager** or connect an Android phone with USB debugging turned on.
6. Press the green **Run** button.

If the app crashes on first launch after a database change, uninstall it from the emulator and run it again so the database is recreated.

Quick test of the strict rule

1. Open the **Pantry** tab and add bread (2, pcs) and cheese (60, g).
2. Open the **Recipes** tab. Cheese Toastie is not in the main list. It appears under **Almost There**, missing butter.
3. Add butter (10, g). Cheese Toastie moves into the main suggestions list.
4. Delete the butter. The recipe disappears from the main list again.
5. Close the app completely and reopen it. Your pantry is still there.

Author

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